

Ganga, GAP, and lockdown: potential threats to the biodiversity of the river

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ABSTRACT

The Ganges River holds a vital status within India as a cornerstone of the nation's ecological and socio-economic fabric. Flowing through diverse terrains like the Himalayas, Gangetic plains, and coastal regions, it fosters an incredibly rich biodiversity, supporting a myriad of life forms ranging from microorganisms to mammals. Beyond its ecological significance, the river sustains the livelihoods of millions, contributing significantly to the economy of northern India. However, rampant human development over recent decades has posed grave threats to its well-being. Pollution from sewage and industrial waste has tainted its waters with heavy metals, posing risks of bioaccumulation and magnification within aquatic life. Infrastructure projects like dams and bridges disrupt natural habitats and impede the migration of aquatic species, further jeopardising the river's ecological balance. Despite these challenges, there have been glimmers of hope, particularly noted during the COVID-19 lockdown when industrial activities ceased, offering the river a temporary reprieve. Moving forward, effective measures are imperative, including reducing industrial and domestic effluent discharge, scrutinising sources of heavy metal contamination, and prioritising the restoration of the river's natural flow. Preserving the Ganges' ecological integrity is paramount, necessitating concerted efforts to ensure its health and vitality for future generations.

Key words: COVID lockdown, Ganga Action Plan, pollution, River Ganga

HIGHLIGHTS

- Role of the River Ganga in Indian socio-economy is immense.
- A great portion of the biodiversity is harboured in the Ganges River.
- Pollution in the river is becoming one of the challenging threats.
- Ganga Action Plan and other recovery measures are not satisfactory.
- COVID-19 lockdown supported the recovery of river conditions.

INTRODUCTION

The Ganga is a major river in the Indian subcontinent, providing livelihoods to millions of people. It originates in the Himalayas and flows eastward across a vast plain to the Bay of Bengal (Figure 1). The Bhagirathi River, which begins at Gaumukh near the Gangotri and Khatilang glaciers in Uttarakhand, is considered the source of the Ganga. The Ganga forms at the confluence of the Bhagirathi and Alaknanda rivers at Devprayag. Spanning about 2,510 km (1,560 miles), it passes through the Indian states of Uttarakhand, Uttar Pradesh, Bihar, and West Bengal. The Ganga's drainage basin covers approximately 861,404 km², accounting for a quarter of India's geographical area (GRBMP 2012). Ganga's major tributaries from the Himalayas' hill or terai regions include the Yamuna, Ghaghara, Gandak, Kosi, Burhi Gandak, and Gomati rivers. Significant tributaries from the Central India plateau include the Damodar, Tons, and Sone rivers.

The Ganga River basin is one of the most productive ecosystems, comparable to the Brahmaputra River basin, another major river system in India. However, there is a great degree of complexity in the management of water and sediment due to the basin's high level of diversity, which includes the Himalayan mountains in the north, deserts in the west, heavily inhabited plains in the centre, and deltas in the east (Singhal *et al.* 2024). Like the Ganga, the Brahmaputra basin is characterised by high variability in discharge and sediment load. Approximately 36.11% of the Brahmaputra basin is located within India. Both river basins are supported by numerous tributaries and are fed by waters from the Himalayas. However, a major difference between them is that the Brahmaputra basin features a complex network of channels, whereas the Ganga's channel network is much simpler. Additionally, the Brahmaputra basin covers a smaller area compared to the Ganga basin

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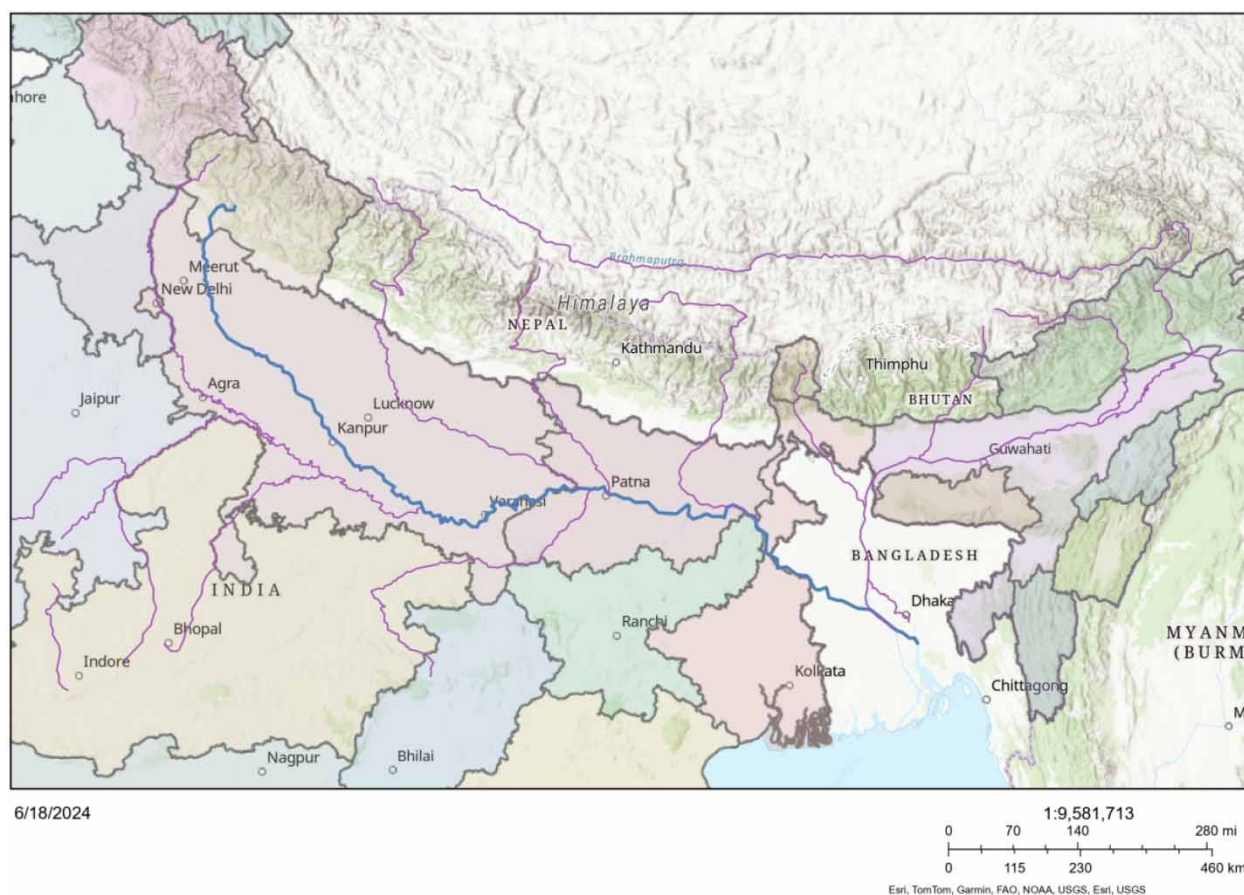


Figure 1 | Map displaying the stretch of the River Ganga with major city and tributaries.

(Sarker *et al.* 2023). These rivers provide a constant source of irrigation and help recharge groundwater along their course. The Ganga significantly supports the livelihoods of about one-third of India's population through agriculture, fisheries, tourism, river-based trade, and transport. The Ganga River basin also holds substantial biological and ecological significance, supporting a wide variety of life forms, from phytoplankton to zooplankton, including the Ganga river dolphin, gharial, otters, and turtles (Sinha & Ahmed 2014). Although all river networks are dendritic structures that are geographically dispersed and serve as channels for runoff, streamflow, sediment transport, and other mechanisms. In recent years, these river networks, specifically Ganga, have been subjected to a continuous stream of external disturbances as a result of climate change, tectonics, erosion, human interference, pollution, and patterns of precipitation (Singhal *et al.* 2024).

Critical Nodes, also known as CNs, are places that have a significant role in facilitating the movement of water and sediment. A total of 207 of these nodes were found in the Gangetic basin, with the bulk of them being located in the northern section of the basin, according to Singhal *et al.* (2024). Within this northern area, 142 projects have been discovered in the upper stretch of the Ganga. These projects include operating projects with a capacity of 4,987 MW and projects that are currently under development with a capacity of 1,751 MW (Dutta *et al.* 2020). These new changes provide the ecology with a number of serious hazards. Hydrological and hydraulic flow regimes may be altered by dams, which in turn can have an impact on the environment and the biodiversity of the area. There is a clear correlation between spatial synchronisation and the long-term survival of species and the stability of ecosystems. As a result of the network structure of the river and the unidirectional flow of water, nutrients, and sediments, asynchronous population dynamics are promoted, which in turn increases the persistence of species. Additionally, the degree of synchronisation that exists across species can be directly impacted by the intricacy of the branching of the river network. The vulnerability of river ecosystems and biodiversity is consequently increased as a result of the construction of more dams, particularly in close proximity to important nodes (Gao *et al.* 2022).

The Ganga is the holiest river in India, with major ancient cities like Haridwar, Kanpur, Allahabad, Varanasi, Patna, and Kolkata situated along its banks. Revered as a mother goddess, the Ganga is believed to purify the soul. Researchers have reported its antimicrobial and medicinal properties (Nautiyal 2009; Rai 2013). For instance, Behera *et al.* (2020) discovered around 69 beneficial bacteria in the sediments of the Ganga and Yamuna rivers. However, pollution in the Ganga is a serious concern, as regular consumption or bathing in its water may cause severe health issues, including cancer (Jain *et al.* 2013). This pollution is primarily due to industrial and sewage discharge from populous cities like Kanpur, Allahabad, Varanasi, Patna, and Kolkata (Paul 2017; Dwivedi *et al.* 2018). Several sewage treatment plants (STPs) are operational and many more are planned to curb the sewage and industrial pollution in the River Ganga (Figure 2). Although these STPs have significant contributions in reducing river pollution, toxic elements, due to pollution, have accumulated in aquatic species such as fish, posing serious health risks to those consuming fish from the Ganga basin. This is evident from the increased incidence of cancer, particularly gall bladder cancer, among the local population (Jain *et al.* 2013; Saini *et al.* 2015).

Regarding the pollution of the Ganga, the Central Pollution Control Board has primarily monitored water quality based on physico-chemical characteristics and faecal coliform counts. However, heavy metal pollution from industrial activities is also a significant concern (Jain *et al.* 2013). The Government of India has implemented several initiatives, such as the Ganga Action Plan (GAP), National Ganga River Basin Authority (NGRBA), and Namami Gange, to address the river's pollution. Despite the expenditure of millions of rupees, there has been no substantial improvement in the Ganga's water quality (Chaudhary & Walker 2019). Conversely, the water quality significantly improved during the COVID-19 pandemic lockdown (Dutta *et al.* 2020), indicating that further measures could lead to significant improvements. This review article deliberates the impact of socio-economic factors and pollution on the Ganga's biodiversity, identifies the sources of pollution, and evaluates

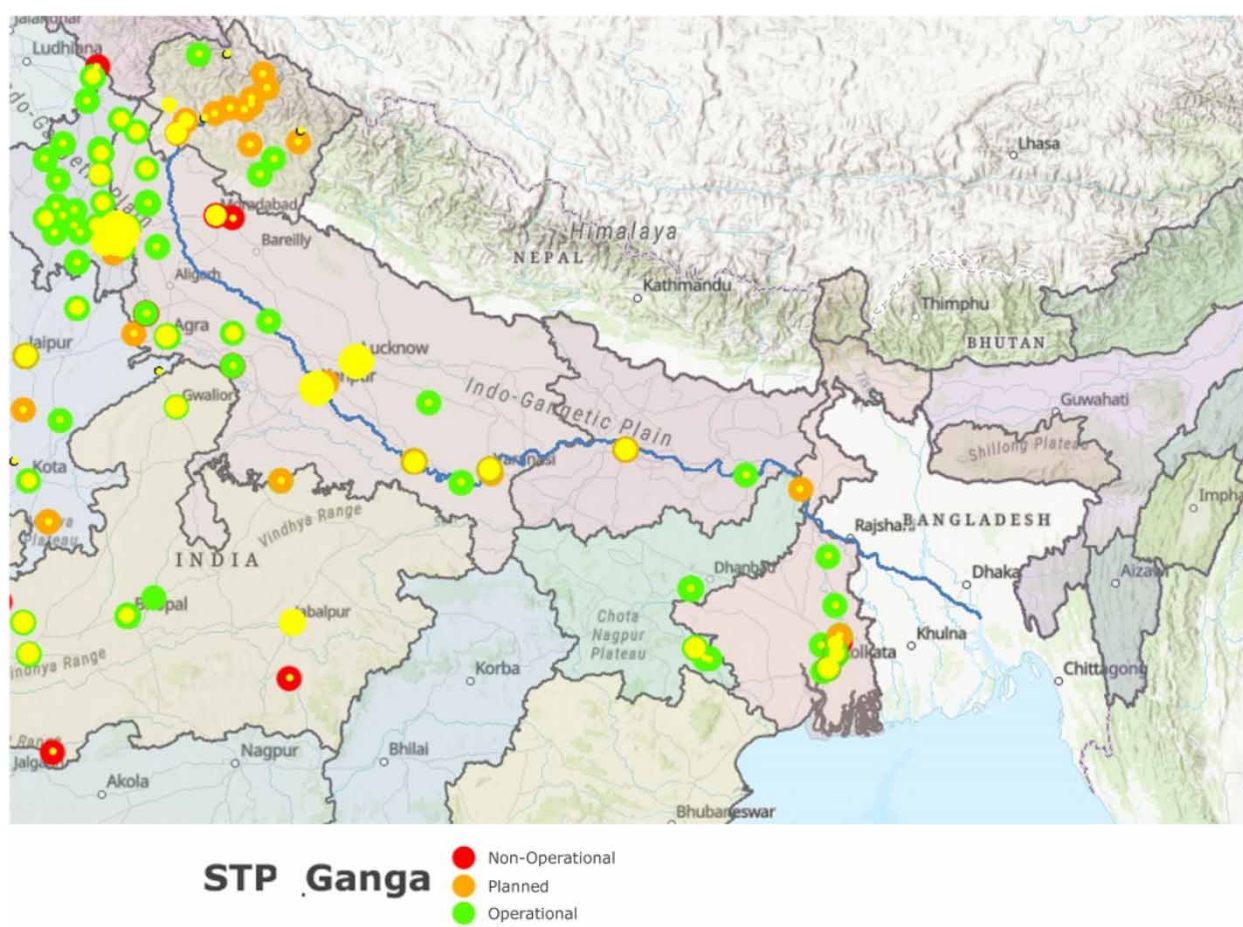


Figure 2 | Sewage treatment plant along the Ganga River basin during the Ganga rejuvenation period. (Source: ArcGIS, https://services.arcgis.com/nUFb6iiYleBwvx5/arcgis/rest/services/STP_Ganga/FeatureServer/.)

the successes and limitations of government initiatives. Additionally, it highlights the impact of the COVID-19 lockdown on the Ganga's pollution levels and proposes future strategies to mitigate pollution.

METHODOLOGY

This review aims to analyse historical data on water quality in the Ganga River and compare it with the quality observed during the COVID-19 lockdown period. The analysis also evaluates the threats to the biodiversity of the Ganga River. The data used for this analysis were collected from available published literature and government project reports, which can be accessed through government websites. Biodiversity status data were obtained from the public literature. Literature was sourced from internet platforms such as Google Scholar and ResearchGate, while pertinent offline publications were retrieved from the library of the Zoological Survey of India. No articles that include important information for the study were excluded. A comprehensive analysis was conducted on a total of 108 pieces of literature and reports for this review. The precise search phrases employed are 'effect of COVID-19 on biodiversity,' 'limnological status of the Ganga river,' 'status of biodiversity in the Ganga,' 'Gangetic plains,' 'effect of Ganga action plan,' and 'pollution-affected Ganga.'

BIODIVERSITY OF THE GANGA RIVER

The Ganga Basin is highly rich in biodiversity and is home to a wide variety of relic, rare, and threatened species (WII-GACMC 2017, 2018; WII-NMCG 2019). For example, the national aquatic animal, i.e. the Gangetic dolphin (*Platanista gangetica*) is found in the River Ganga. Three species of otters, viz. the smooth-coated otter (*Lutrogale perspicillata*), Eurasian otter (*Lutra lutra*), and the small-clawed otter (*Aonyx cinereus*) are also found in the River Ganga. The only surviving critically endangered Gharial species (*Gavialis gangeticus*) is found in the River Ganga. Furthermore, two species of Mugger, i.e. Indian Marsh crocodiles (*Crocodylu spalustris*) and Estuarine crocodiles (*Crocodylus porosus*) are found in the River Ganga and its tributaries. The River Ganga is considered a global hotspot for freshwater turtle diversity and among 24 species of freshwater turtle found worldwide, 13 species are found in the River Ganga including the critically endangered *Batagur kachuga*. Among 13 species of freshwater turtles, 9 are hard-shell and 4 are soft-shell turtles. Amphibians are considered secondary consumers in the food chain and play a vital role in both terrestrial and aquatic ecosystems. Around 49 species of amphibians are found in the River Ganga basin. About 150–200 species of fish are found in the River Ganga, which includes the critically endangered Ganges shark (*Glyphis gangeticus*), Gangetic stingray (*Himantura fluviatilis*), Golden mahseer (*Tor putitora*), and Hilsa (*Tenualosa ilisha*). Among 1,340 species of birds found in India, 177 species of migratory and resident-breeding birds are found in Gangetic plains, e.g. mynah, parrots, crows, kites, partridges, and fowls. There are eight species of resident-breeding birds. Among eight resident-breeding birds, one species is listed as endangered and one in the vulnerable category, whereas three are nearly threatened as per the IUCN Red List. The Ganga river basin is also rich in floral diversity and around 289 plant species belonging to 88 families and 219 genera are found in the basin (WII-NMCG 2019). These include 56% herbs, 10% shrubs, 21% trees, 6% climbers, and 75 grasses. The *Asteraceae* is the dominant family, whereas *Cynodon dactylon* is the prominent species found in Gangetic plains.

Wildlife Institute of India has consolidated the baseline data of the aquatic species of the River Ganga and divided the river into six biodiversity zones (WII-GACMC 2017). The aim was to identify the areas with high biodiversity values and threats to the ecological variety of the river Ganga. These six stretches of high biodiversity value cover approximately 50% length of the River Ganga.

1. **Devprayag to Rishikesh (Uttarakhand):** This high biodiversity stretch is 61 km long and nurtures 93 phytoplankton, 76 periphyton, and 19 zoobenthic species. About 56 species of fish are recorded from this stretch. The endangered golden mahaseer and the critically endangered snow trout are also found in this stretch. The smooth-coated and Eurasian otters have also been reported from this stretch.
2. **Makdumpur to Narora (Uttar Pradesh):** This stretch is 147 km long and is characterised by a low depth of river (1.5–5.0 m). The stretch from Brijghat to Narora is considered a Wetland of International Importance. More than 80 fish species, about 62 water birds, 35 phytoplankton, and 31 zooplankton have been reported from this stretch. It is interesting to note that 110 km of this stretch is protected by Hastinapur Wildlife Sanctuary (established in 1986). Swamp Deer (*Rucervus duvaucelii duvaucelii*), Hog deer (*Axis porcinus*), Blackbuck (*Antelope cervicapra*), Nilgai (*Boselaphus tragocamelus*), Wild boar (*Sus scrofa*), Golden jackal (*Canis aureus*), Jungle cat (*Felis chaus*), and Fishing cat (*Prionailurus viverrinus*) are some common mammalian species found in Hastinapur Wildlife Sanctuary. This sanctuary also supports

180 species of birds including migratory water birds visiting the area during winter. Asian Openbill (*Anastomus oscitans*) and Sarus crane (*Grus antigone*) are also recorded as breeding here.

3. **Bhithaura to Ghazipur (Uttar Pradesh):** This 454 km stretch is the longest stretch with high biodiversity. This stretch supports a total of 50 fish, 357 phytoplankton, 42 zooplankton, 19 periphyton, and 45 zoobenthic species. This zone also includes Kachhua Wildlife Sanctuary (established in 1989) from Rajghat to Ramnagar in Varanasi. Turtles, Ganges dolphin, and other water animals, such as rohu, bhakur, tengra, prawn, nain, and bam can be found here. The rarest of the rare species of tortoises like Sundari kachhua, Pechari, and Kachhua Tongukaare are found in this sanctuary.
4. **Chhapra to Kahalgaon (Bihar):** About 61 fish and more than 80 phytoplankton and zooplankton species are recorded from this 296 km long stretch. Furthermore, 65 km of this stretch from Sultanganj to Kahalgaon is protected as Vikramshila Gangetic Dolphin Sanctuary as notified in 1991. Ganga river dolphins (*Platanista gangetica gangetica*), Indian smooth-coated otter (*L. perspicillata*), gharial (*G. gangeticus*), and a variety of freshwater turtles are found in this sanctuary.
5. **Sahibganj to Rajmahal (Jharkhand):** This stretch of the River Ganga is only 34 km long, however, it nurtures about 89 fish, 182 phytoplankton, and 40 zooplankton species.
6. **Baharampur to Barrackpore (West Bengal):** This 246 km long stretch of the River Ganga is the habitat of 25 fish, 44 phytoplankton, and 21 zooplankton species.

THREATS TO THE BIODIVERSITY OF THE RIVER GANGA

Gangetic plains are characterised as one of the 10 biogeographic zones of India due to its rich biodiversity (Singh & Chaturvedi 2017). Nilgai, Blackbuck, Chinkara, Swamp deer, Hog deer, Rhino, Bengal florican, Hispid hare, Sambar, and Chital are some important wildlife of the Ganga biogeographic zone. However, the biodiversity of the Ganga basin, especially the aquatic life of the River Ganga, is at risk due to habitat degradation. The reason behind the variations in biota diversity and community structure is primarily related to hydrological alterations due to dam constructions, reduction in water level due to irrigation, land erosion due to change in land use pattern, rapid sedimentation due to deforestation, exotic species invasion, climatic changes, pollution, and over-exploitation of riverine resources (Sarkar *et al.* 2012). A series of barrages and dams from Rishikesh to Narora including the Tehri dam constructed in the hills of Uttarakhand has considerably reduced the water flow. Therefore, species like dolphins and gharials are nearly sightless and considered endangered because they largely depend on water depth and prefer more than 4 m depth. Freshwater turtle populations are declining in the River Ganga due to toxic substances released from agricultural fields and industries. The alteration in the natural flow of rivers, due to dams and barrages constriction, has also affected their population. Sand mining has degraded the nesting habitats of turtles, therefore largely affecting the decline in their population (Stanford 2020). It also has adverse impacts on aquatic species, particularly fish, which are declining rapidly from its natural habitat. Thus, urbanisation and chemical and physical alterations of natural habitats due to the growing number of hydroelectric and irrigation projects are some of the reasons behind the degradation of biodiversity (Dorber *et al.* 2020). Some of the factors responsible for the degradation of the biodiversity of rivers and their effect on human beings are discussed briefly.

Socio-economic factors threatening the biodiversity of the River Ganga

During the Vedic age, Indus and Sarasvati were the major holy rivers in India. However, climatic change and the fading of the Sarasvati river along with other elements resulted in the fall of the Harappa civilisation in the Indus Valley. Thus, the centre of Indian civilisation shifted from the Indus basin to the far lush Ganga River basin in the early second millennium BC (McIntosh 2008). From the Maurya to the Mughal Empire, the Ganga River basin was the centre of successive dominant states. Therefore, the river Ganga has enormous cultural and economic significance in India. However, some cultural and economic activities are threatening the existence of the river and some of them are listed in the following.

Threats due to religious activities

Some important pilgrim centres, such as Rishikesh, Haridwar, Garhmukteshwar, Kannauj, Allahabad, Mirzapur, Varanasi, and Gangasagar are situated at the bank of the river, where various religious events take place throughout the year. Ritual bathing in the River Ganga is thought to purify the body and soul, therefore millions of people bathe at a specific stretch of the river during Kumbh (Figure 3) (Tyagi *et al.* 2013). In the 2013 Kumbh at Prayagraj, about 88.7 million people took a bath in the River Ganga within a 45-day event. The biggest gathering in human history was on 10 February 2013 when 30 million people took a holy dip in a single day (Rani *et al.* 2014). A significant enhancement of *Salmonella typhi* in the

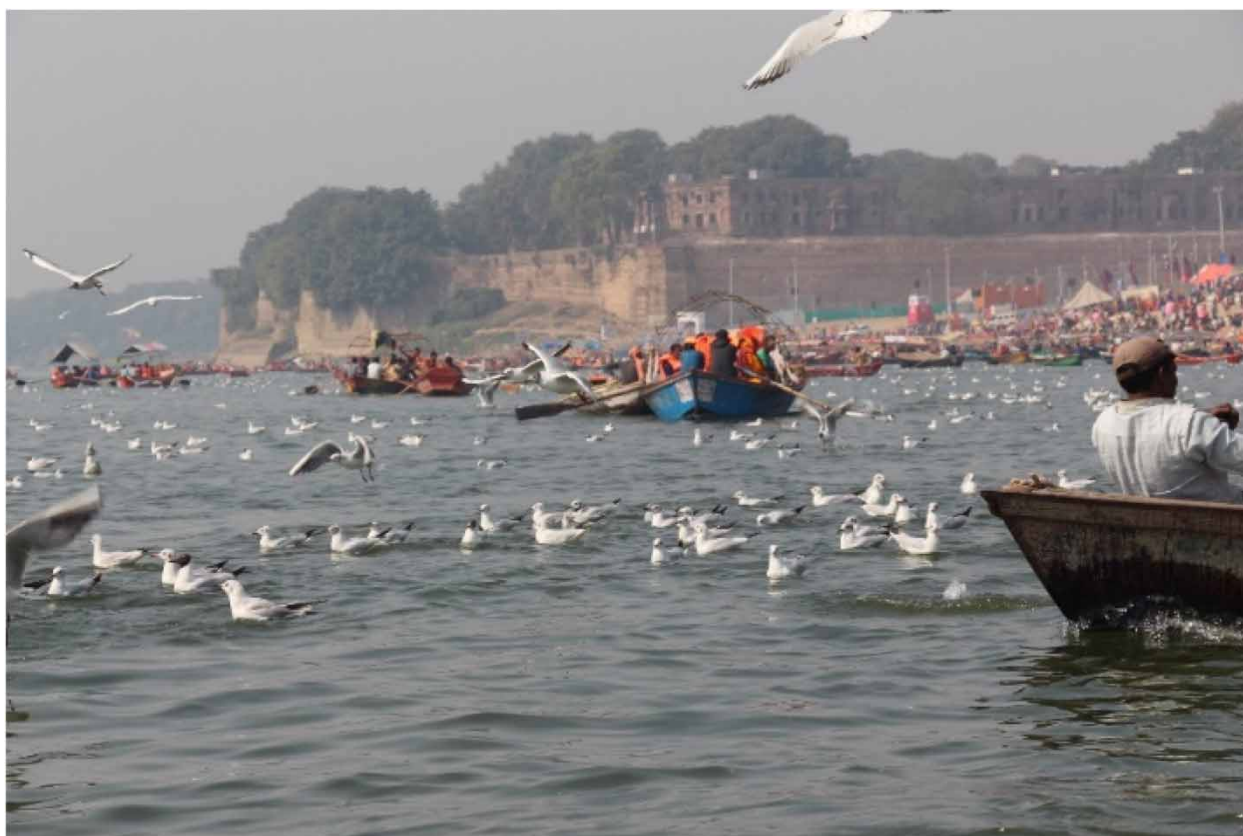


Figure 3 | Kumbh-2019 at Sangam, Prayagraj, Uttar Pradesh, India.

water and sediment of the River Ganga was reported during Kumbh 2013 at Prayagraj. Srivastava *et al.* found that the water quality of the river deteriorated during mass bathing during Kumbh 2013 at Prayagraj (Srivastava *et al.* 2013; Rani *et al.* 2014). Besides Kumbh, several other bathing fairs deteriorate the water quality of the River Ganga, such as Somwati Amavasya, Kartik Purnima, Shravan Amavasya, and Ganga Dussehra. The deterioration in water quality is mainly attributed to the use of soaps, shampoos, detergents, polythene, food, flowers, leaves, milk, curd, ghee, coins, etc. (Khanna *et al.* 2012). Vortmann *et al.* (2015) have found in their study that the Ganga at Prayagraj is unsafe for bathing and can cause serious health problems due to extremely high levels of coliform bacteria. It is found that in the year 1817 Kumbh played an important role in the spread of Cholera and the subsequent global pandemic (1817–1824) (Banks 1961; Bryceson 1977). Cholera outbreak in the year 1948 Kumbh resulted in mass vaccination of people from Cholera for the year 1960 event. Similarly, the COVID-19 pandemic resulted in the mass infection in Kumbh 2021 at Haridwar. Therefore, mass bathing at the bank of the River Ganga has not only deteriorated the water quality but also resulted in the spread of infectious diseases.

Idol immersion is another religious activity which is polluting the River Ganga. For example, idol immersion during Durga Pooja results in a significant increase in temperature, pH, conductivity, BOD, COD, total alkalinity, chloride, total hardness, and phosphate of the River Ganga in West Bengal (Sarkar 2013). Similarly, Chaturvedi & Pandey (2006) found an increase in the concentration of heavy metals, such as Zn, Fe, Ni, Cu, Cr, and Cd in the River Ganga at Vindhyachal Ghat during Navratri Pooja. It was found that during the idol immersion, a lot of vermilion is thrown into the river which increases the levels of Pb and Cr (Das *et al.* 2012). Furthermore, the cremation of dead bodies and the release of their ashes in rivers is also a threat to the water quality and biodiversity of the river. Tripathi & Tripathi (2014) have reported that on average 32,000 dead bodies were cremated every year between 2009 and 2011 at Harishchandra and Manikarnika Ghat, in Varanasi. Therefore, it would be advisable to stop the immersion of idols, flowers, and dead bodies in the river to protect its water from deterioration.

Threats due to economic activities

Economic activities, such as power projects, irrigation, and waterways have a tremendous effect on the growth of the Indian economy, however, these activities have also threatened the existence and well-being of the river. Here, some concern related to economic activity on the biodiversity of rivers is discussed briefly.

- a. Hydroelectric power:** The Ganga basin has been assessed having a 20,711 MW hydroelectric potential (Attri *et al.* 2017). Uttarakhand is an ideal place for hydropower development due to the presence of mighty rivers like Ganga, Yamuna, Kosi, etc. The Uttarakhand power department has planned several projects on its rivers, such as Bhagirathi and Alaknanda, which will affect the river flow through diversion to tunnels or making reservoirs. Therefore, the Wildlife Institute of India studied the environmental impacts of 70 hydroelectric projects on the terrestrial and aquatic biodiversity of the Bhagirathi and Alaknanda riverine ecosystems (Rajvanshi *et al.* 2012). They have found that the hydropower projects in Uttarakhand are threatening the biodiversity of the River Ganga, especially migrant species. There are around 17 migrant fish species found in Alaknanda and Bhagirathi basins including three species of long-distance migrant Mahseer. Therefore, the construction of a dam or a barrage across a river will not only hamper their migratory behaviour but also affect their breeding cycle. It is evident from Tehri dam that their migration has already been obstructed. Even the wildlife of Sunderban has been affected by the construction of dams upstream. These constructions have reduced the freshwater delivery in Sunderban, and hence, there is saltwater intrusion and loss of islands. Furthermore, the glacial disaster in 2020 in the Chamoli district of Uttarakhand sends a strong message to rethink the model of development being followed, particularly with regard to the exploitation of its hydroelectric potential.
- b. Irrigation:** Use of the Ganga's water for irrigation has been common since ancient times as described by the Greek historian Megasthenes, who visited India during the reign of the emperor Chandragupta Maurya, in the 4th century B.C. (Singh 2017). In modern India, several dams and barrages were constructed on the River Ganga for irrigation and power generation. For example, the Bhimgoda barrage at Haridwar diverts a large amount of water into the upper Ganga canal in order to provide water for irrigation. During monsoon months, the barrage at Bijnor in Uttar Pradesh diverts water into the middle Ganga canal. Similarly, Narora Barrage in Uttar Pradesh further diverts the water into the lower Ganga canal. The Farakka barrage in West Bengal controls the flow of the river to make the river Hooghly navigable and make Diamond Harbour functional. There are around 30 diversions on the main stem and tributaries of the River Ganga and a majority of them are in Uttar Pradesh. The cultivated areas of Uttar Pradesh and Bihar benefited from the vein of irrigation canals. These irrigation canals are responsible for the increase in the production of sugarcane, cotton, and oilseeds in Gangetic plains. However, these irrigation canals divert over 60% of the Ganga's annual flow and leave very little water to flow in the river during dry months (Shah & Rajan 2019). In Uttarakhand, the hydroelectric power generation at the Tehri dam on Bhagirathi further regulates the downstream flow of the River Ganga. Mukherjee *et al.* (2018) have found in their study that the over-abstraction of groundwater in the Ganga basin has lowered the flow of the river and adversely affected the water quality of the river. This deterioration in the water quality of the river due to the reduction in the water flow is also a threat to aquatic life. Therefore, environmental flows of rivers should be maintained throughout the year to sustain freshwater and estuarine ecosystems as well as human livelihoods.
- c. Navigation:** According to Megasthenes, the Ganges and its main tributaries were important transportation routes in ancient times (Nagabhatla & Jain 2013). In the 19th century, paddle steamers revolutionised inland transport where irrigation-cum-navigation canals formed the main channels of the water transport system. Regular steamer services were operational from Kolkata to Prayagraj and far beyond. In the mid-19th century, the construction of railways has declined water transport in the River Ganga and several other rivers. Navigation in the River Ganga is further affected by the increasing withdrawal of water for irrigation. At present, navigation in the River Ganga is insignificant and only West Bengal is using waterways to transport agricultural and rural products, such as jute, tea, and grain. In other parts, such as Varanasi it is used for tourism purposes only, especially for Ganga Arti (Figure 4(a)). In order to regulate inland waterways, the Government of India is constructing approximately 1,000 miles of National Waterway 1 from Prayagraj to Haldia in the River Ganges to facilitate trans-boundary trade relations with Nepal and Bangladesh (Figure 4(b)).

However, the adverse effect of these waterways on aquatic biodiversity cannot be ignored. For example, Gangetic Dolphin is essentially a blind animal, which navigates and catches its prey by sound using echolocation. Therefore, the movement of large vessels affects the survival of dolphins in the river. Activities like dredging to maintain the deep-water channel for

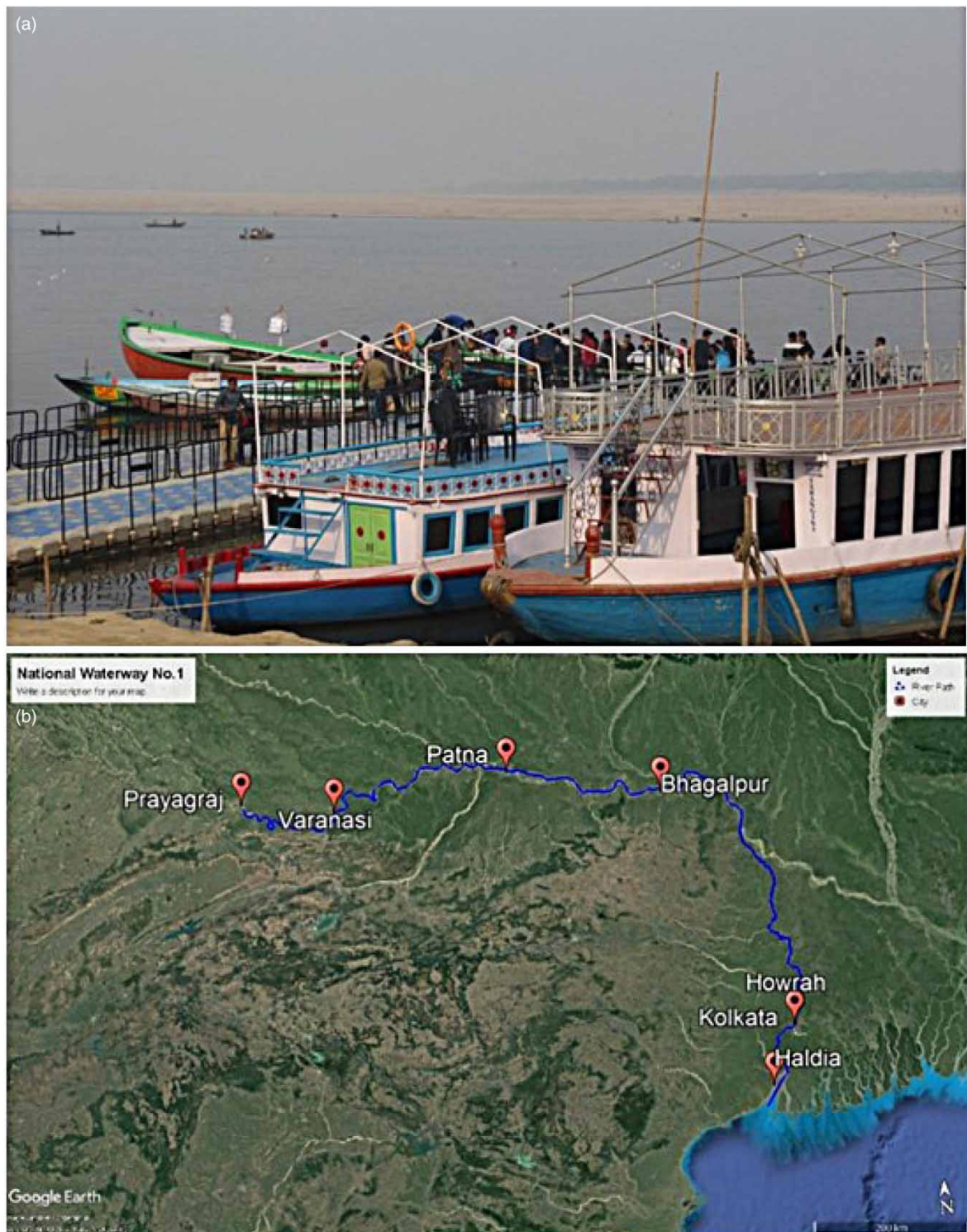


Figure 4 | (a) Navigation in the River Ganga at Varanasi, Uttar Pradesh and (b) National Waterway 1 between Prayagraj and Haldia.

navigation will cause tremendous destruction to aquatic animals, such as fish and turtles, and their habitat. Dredging of shallow areas will disturb their habitat as well as population, as they lay their eggs in river beds. Thus, National Waterway 1 in the present scenario appears unfeasible, as the benefit from it to the national economy will be at the cost of biodiversity loss. If we

consider the associated environmental and social damage due to the alteration of the river and the development of infrastructures then it appears to be an economic disadvantage.

Pollution threats to biodiversity of the River Ganga

Most of the aquatic organisms are very sensitive to their environment and respond in different ways to the pollution. These include either death or migration to any other habitat, reduction in reproductive capacity or suppression of some enzyme systems needed to conventional metabolism (Deborah 1996). The exponential growth of population, industrialisation, urbanisation, and rising standards of living in Gangetic plains have exposed the River Ganga to various forms of degradation. It has been found that 90% of the river's natural discharge is extracted through dam and irrigation when the river reaches Kanpur (Vega *et al.* 1998). Sewage pollution, industrial effluent, agricultural runoff, and religious activities are the four main sources of pollution in rivers, where municipal sewage is the main contributor followed by industrial effluents. Various organic and inorganic substances originating from these sectors are responsible for the pollution of rivers. Pollution due to religious activity is already discussed in the previous part, therefore effects of sewage, industrial effluent, and agricultural runoff are discussed in this section.

1. Sewage pollution: India has witnessed rapid urbanisation in recent decades as shown in Figure 5, where urban area increased several fold. Urbanisation at such a scale is responsible for environmental problems such as water supply, wastewater generation, treatment and disposal. Kanpur, Prayagraj, Varanasi, Patna, and Kolkata are some major cities situated on the bank of the River Ganga. Although there are several STPs functioning in these cities (Figure 2), the water quality of the river Ganga is rapidly deteriorating due to the discharge of untreated sewage from these urban centres. Sewage water constitutes heavy metals (Cr, As, Cd, etc.), organic matter (pesticides, detergent, etc.), nutrients (viz. N, P, K), inorganic matter (Ca, Mg, Na, K, etc.), and pathogens (bacteria, fungus, etc.) (Vega *et al.* 1998). Das (2011) reported that 75% pollution of the River Ganga is related to the discharge of millions of litres of sewage generated per day in the towns along the River Ganga. According to CPCB's recent report, about 8,250 MLD (million litre per day) of wastewater is generated from 223 cities/towns along the River Ganga, out of which about 2,460 MLD is directly discharged into the River Ganga (Shukla *et al.* 2021). It is important to note that

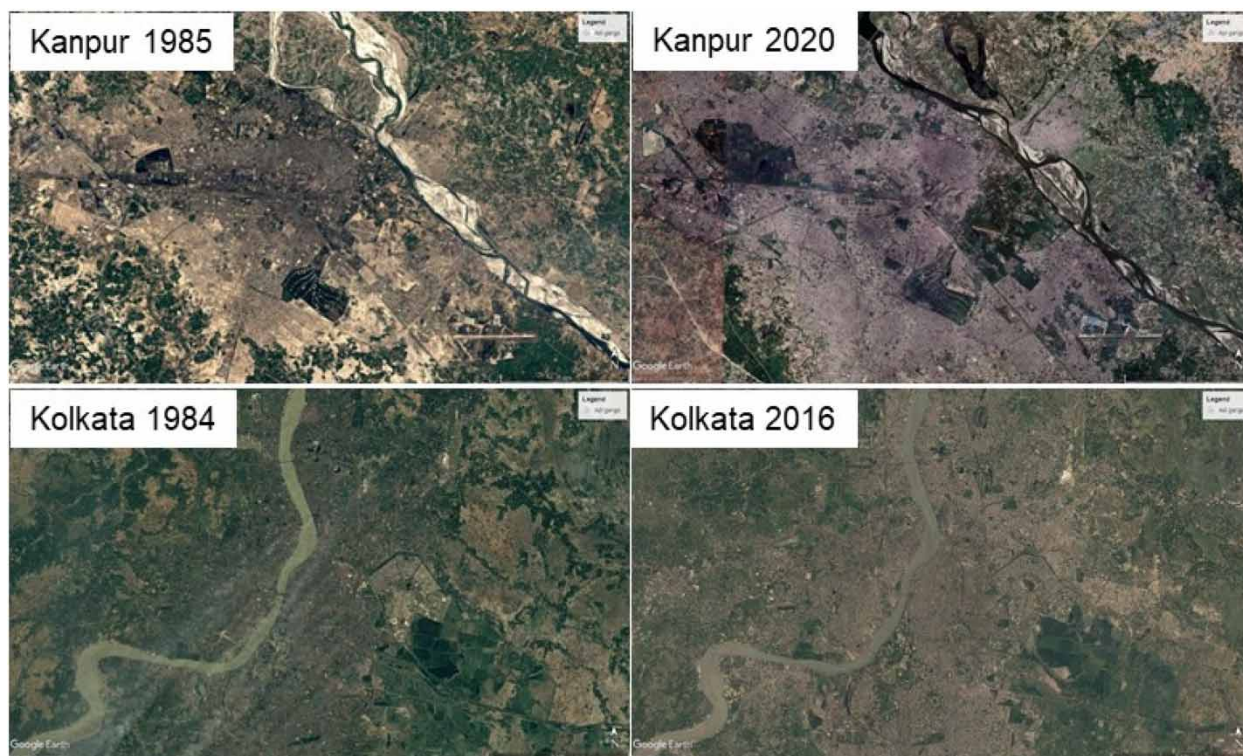


Figure 5 | Satellite images of Kanpur and Kolkata displaying the change in urbanisation in nearly 30 years.

this untreated sewage water is not only polluting rivers but is also contaminating groundwater due to percolation. Furthermore, municipal water treatment facilities generally do not remove traces of heavy metals which is also a concern for the riverine ecosystem. Therefore, aquatic life as well as people living along the River Ganga are exposed to unknown quantities of pollutants. It is interesting to note that the water quality of the River Ganga, instead of improving, further deteriorated at many stretches during the 68-day COVID-19 pandemic lockdown from March 25, 2020 (Dutta *et al.* 2020; Shukla *et al.* 2021). This was mainly due to the discharge of untreated or partially treated sewage in the river, as curbs on human activities have almost no effect on sewage generation. Sewage pollution severely degrades riverine ecosystems and exacerbates serious risks to public health. For example, alkylphenolpolyethoxylates or phthalates present in sewage effluents due to degradation of detergents may cause turbulence to the endocrine systems of male fish by setting off varying levels of oestrogen-mimetic activity (Jobling *et al.* 1996; Tyler *et al.* 1996). Furthermore, an increase in the concentration of phosphates, nitrates, and other organic wastes in water bodies will result in eutrophication. This will result in excessive growth of algae and bacteria and increase the BOD level, which will consequently hamper aquatic animal life.

2. **Industrial pollution:** Industrial pollutants containing heavy metals are a major concern for healthy riverine ecosystems. Dutta *et al.* estimated that around 20% of the total volume of wastewater discharged in the River Ganga is from industrial sources (Dutta *et al.* 2020). Approximately 1,072 grossly polluting industries (GPIs), mostly situated in Uttar Pradesh are directly discharging their effluent into the River Ganga (Namami Gange 2020). Kanpur (tanneries), Prayagraj (engineering), and Varanasi (carpets and locomotive) are the key industrial cities situated at the bank of the River Ganga in Uttar Pradesh. Therefore, the area from Kannauj to Varanasi is considered the most polluted stretch of the River Ganga. The tannery sector contributes considerably to the pollution of the River Ganga, as it accounts for the highest proportion of GPIs. Besides the tanneries, other industrial effluents, such as textile, paper mills, and sugar mills are also significantly enhancing the concentration of toxic elements in the River Ganga (Bhattacharya *et al.* 2008; Aktar *et al.* 2010; Rai *et al.* 2010). The industrial waste includes hydrochloric acid, bleaches, dyes, pesticides, and other toxic heavy metals (As, Cd, Cr, Hg, Ni, and Pb). It has been reported in literature that the levels of carcinogenic elements like As, Cd, Cr, Hg, Ni, and Pb were exceeding the WHO permissible limit for potable water at many sites of the Ganga River (Sinha *et al.* 2007; Aktar *et al.* 2010; Pandey *et al.* 2010; Katiyar 2011; Pal *et al.* 2012). These substances gradually enter the food chain and start accumulating as toxins and those animals which are higher up in the food chain accumulate much more of this toxin in their body. It has been found in a recent study by Sen and co-workers that the concentration of dissolved heavy metals in the River Ganga was 50% reduced during the nationwide COVID-19 pandemic lockdown (Shukla *et al.* 2021). It was attributed to either no discharge or low discharge of industrial effluents as most of the industries were closed in a given period of time. Furthermore, the flow of the river was also adequate due to less consumption of river water for industrial and agricultural activities. These results suggest that proper treatment of industrial effluents and restoration of the ecological flow of rivers is necessary to combat industrial pollution.
3. **Agricultural runoff:** The agriculture of northern India basically depends on the River Ganga and her tributaries for irrigation. Nearly one-third of India's population depends on the food provided by the Gangetic plains, therefore, their reliance on the Ganga and her tributaries is inestimable (Pal *et al.* 2009). In the last four decades, the crop output in Gangetic plains has increased tremendously due to crop diversification and improvement in the irrigation facilities (from 134 to 411 thousand hectares per district), which has reduced the dependency of farmers on the monsoon (Tare *et al.* 2013, 2015). Although the cultivable land is limited in the Gangetic plains, the crop production increased approximately four-fold due to the massive use of fertilisers and pesticides, for example, use of fertiliser has also increased massively from 1,700 to 76,300 tonnes per district during the four decades due to the Green Revolution (Tare *et al.* 2015). Similarly, India ranks fourth in the world for pesticide production, producing approximately 85,000 MT of pesticide annually (FICCI 2016). Many of these inputs, such as fertilisers and pesticides are harming the Ganga waters, and her water is harming their crops. Many cancer-causing chemicals, such as DDT or HCH, are sprayed as pesticides in agricultural farms which wash down into the river and thus pollute its water for drinking, bathing, or cooking (Alavanja & Bonner 2012; FICCI 2016). Many pesticides, such as DDT, Aldrin and HCH, are still in agricultural use, although the Government of India has banned their use (Abhilash & Singh 2009; Vijgen *et al.* 2011). Sah *et al.* (2020) in their recent study found 13 banned and restricted organochlorine pesticides (OCPs) in the surface water of the River Ganga. As compared to other parts, the lower stretch of the river was most contaminated with these chemicals. These chemicals have adverse effects on fish production in the basin and statistics suggest that annual production of fish declined from 85.21 tonnes during 1959 to 62.48 tonnes during 2004 (Payne *et al.* 2004; Das 2007). Thus, these chemicals are damaging all forms of aquatic

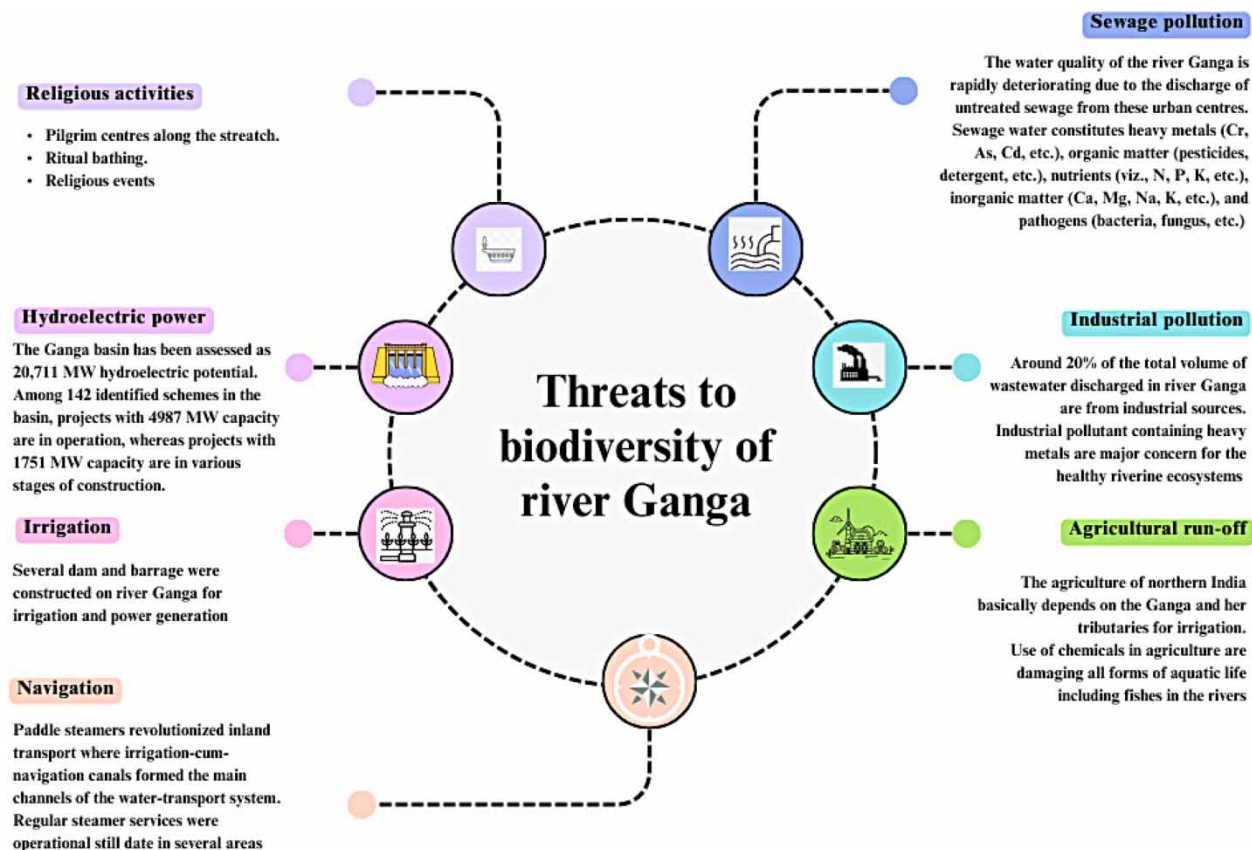


Figure 6 | Different socio-economic factors threatening the biodiversity of the River Ganga.

life including fish in the rivers, which is hampering the livelihood of fishermen. Furthermore, the use of fertilisers and pesticides adversely affects the community structure of insects, populations of soil invertebrates and microorganisms (McLaughlin & Mineau 1995; Oosthoek 2013). Therefore, it is necessary to stop the use of toxic pesticides and fertilisers and use organic farming. Furthermore, most of the the river water is diverted for irrigation, therefore conscious use of water for irrigation is necessary to maintain the ecological flow of the river. Different socio-economic factors threatening the biodiversity of the River Ganga are summarized in Figure 6.

REJUVENATION OF GANGA AND ITS LIMITATIONS

In 1896, British bacteriologist Ernst Hankin reported the antibacterial properties of Ganga water against *Vibrio cholera* (Hankin 1896). Similarly, in the early 20th century, French microbiologist D'Herelles also reported the antibacterial properties of Ganga water (D'Herelle 1917). Recently, metagenome sequence analysis from the sediment of Rivers Ganga and Yamuna revealed the presence of 69 beneficial bacteria in the basin (Figure 7) (Behera *et al.* 2020). In this study, it is also found that the relative abundance of bacterial species is highly related to the different pollution levels. Thus, the river Ganga has characteristic microbial diversity that might have a role in the reported self-cleansing property. However, socio-economic changes in the basin after the post industrialisation era have severely polluted the water of the River Ganga and surpassed its self-cleansing properties (Dwivedi *et al.* 2020). In order to maintain the water quality of the River Ganga for drinking and bathing, the Indian government has taken many measures to clean the river. Here, the success and limitations of some of these cleaning measures of the River Ganga are discussed briefly.

1. GAP: Government of India, under Central Ganga Authority (CGA), initiated GAP I in 1986 in order to curb the pollution of the river. The objective of the GAP I was to restore the water quality of the river to 'Class B' or the 'Bathing-Class'1 (Tare *et al.* 2003). The focus of GAP I was mainly on sewage interception and treatment facilities in order to maintain the BOD at

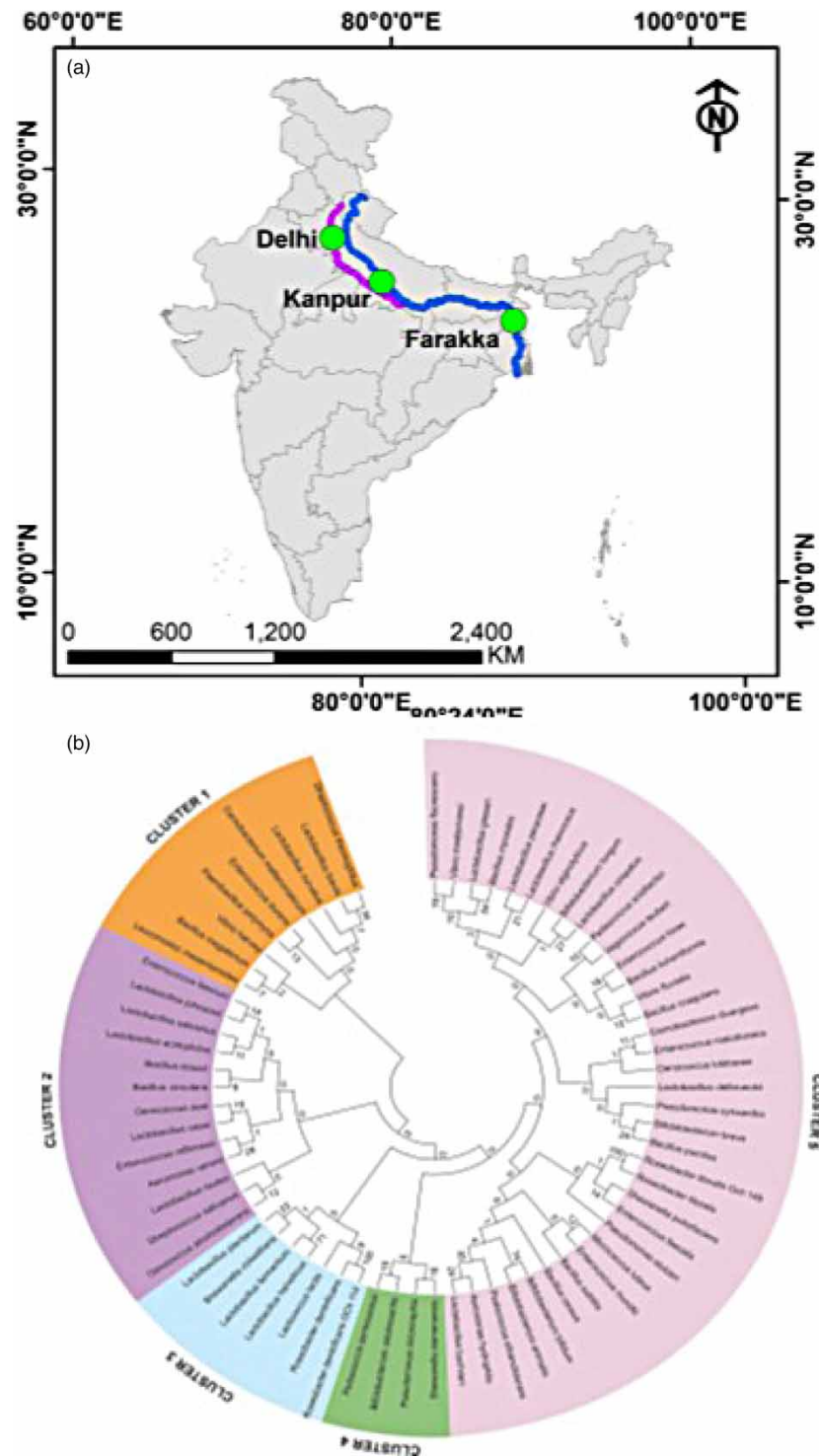


Figure 7 | (a) Map showing the sediment sampling sites and (b) phylogenetic tree of 69 identified genome of helpful bacterial species derived from the sediments metagenome of Ganga and Yamuna Rivers (Behera *et al.* 2020).

3 mg/l, DO at 5 mg/l, total coliform at 10k/100 ml and faecal coliform at 25k/100 ml. However, GAP-1 has little success in checking the pollution of rivers mainly because pollutions of tributaries are ignored. Therefore, CGA introduced the GAP II while GAP I was still in progress. In GAP II, pollution along the three major tributaries (Yamuna, Damodar, and Gomti)

of the River Ganga is targeted including 25 Class I towns with a population of more than 100,000. However, GAP largely failed in curbing the pollution as per a report published in 2000 by the Comptroller and Auditor General of India (CAG 2000). According to this report, GAP achieved only 39% of its target, whereas it has consumed 91% of its total allocated budget. Three reasons were mentioned in this report for the failure of GAP, viz. discrepancies in the selection of cities and towns, inaccurate estimation of sewage load, and variation in the performance of different states. It has been found that GAP focused only on sewage and effluent discharge from major cities and failed to use an integrated river basin management approach (Ostrom 2009). For example, only 45% of industries along the River Ganga have installed effluent treatment plants (ETPs) out of which 18% failed to satisfy recommended standards (Chaudhary & Walker 2019). Furthermore, anaerobic activated sludge blanket technology adopted by GAP was not capable of removing bacterial contamination I (Menon 1988; Tare *et al.* 2011). The strengths, weaknesses, opportunities, and threats (SWOT) of GAP are given in Table 1 (Tare *et al.* 2011).

2. **NGRBA:** GAP focused largely on engineering-based approaches and lacked the involvement of local communities; therefore, desired success was not achieved in restoration of water quality of the River Ganga. In order to maintain the water quality, ecological flows, and biodiversity of the River Ganga, NGRBA was constituted by the Government of India in 2011. NGRBA functions include prevention, control, and abatement of pollution in the River Ganga. Therefore, the National Mission for Clean Ganga (NMCG), an implementing body of NGRBA, has initiated several biodiversity conservation projects to restore the dwindling aquatic wildlife of the river, such as Biodiversity Conservation and Ganga Rejuvenation, Fish and Fishery Conservation in Ganga River, Ganges River Dolphin Conservation Education Programme. In a recent study, it has been found that heavy metal contamination in the River Ganga has relatively declined after the establishment of NGRBA (Dwivedi *et al.* 2018). However, in 2016, NGRBA was replaced by the National Council for Rejuvenation, Protection, and Management of River Ganga and its jurisdiction was extended to the states which comprise the Ganga River Basin, namely, Uttar Pradesh, Himachal Pradesh, Uttarakhand, Bihar, Madhya Pradesh, Chhattisgarh, West Bengal, Rajasthan, Haryana, Jharkhand and Delhi-NCR, and any other states having the major tributaries of the Ganga.
3. **Namami Gange Programme:** The Namami Gange Programme was initiated by the Government of India in June 2014 for the effective abatement of pollution, conservation, and rejuvenation of the River Ganga. The Namami Gange programme is making a major shift in implementation strategies as compared to previous schemes by involving all the concerned states and grassroots level institutions, such as urban local bodies and Panchayati Raj. It has been proposed to increase the capacity of STPs under this programme. Nearly 63 sewerage management projects are under implementation and 12 new sewerage management projects are launched under this programme. Riverfront development programmes have been initiated for the construction, modernisation, and renovation of 182 ghats and 118 crematoria. Real-time water quality monitoring stations have also been installed for the monitoring of industrial effluent. Furthermore, biodiversity conservation and afforestation along river banks have also been introduced for the conservation of dolphins, turtles and different kinds of fish. Therefore, several biodiversity centres (Rishikesh, Dehradun, Narora, Allahabad, Varanasi, Bhagalpur, Shahibganj, and Barrackpore) will be developed along the River Ganga. In a significant move, around 10 crore trees will be planted along the bank of the river from Haridwar to Kolkata. Institutions like the Wildlife Institute of India, Central Inland Fisheries Research Institute and Centre for Environment Education will actively participate in the plantation drive. It has been experienced that cleaning of the River Ganga is almost impossible to achieve without public participation, therefore, numerous awareness activities through rallies, campaigns, exhibitions, etc. have been organised under this programme.

These projects have yielded sub-optimal results and the pollution problem of the River Ganga is still persistent. This is mainly because the capacity of STPs is not increasing in the same proportion as urbanisation is increasing day by day. Apart from town or city, wastewater is also generated from non-point sources, such as agricultural runoff, which is very hard to tackle. It is also important to note that the ecological flow of the river is essential for its good health, but the base flow amount of the River Ganga has sharply reduced by 56% in the last four decades due to the rapid surge in water demand (Mukherjee *et al.* 2018). Over-exploitation of surface and groundwater due to fast urbanisation in Gangetic plains resulted in water scarcity in the River Ganga. Therefore, cleaning of the River Ganga without maintaining its ecological flow is questionable.

Table 1 | SWOT analysis of GAP

Strengths of GAP	Weaknesses of GAP	Opportunities for future	Threats and challenges
Interception and diversion of Nalas – The GAP intercepted Nalas and built a conveyance system with pumping stations to shift sewage flow to STPs before building the network.	Limited scope of issues addressed – GAP failed to identify concerns unrelated to river contamination. For instance, diverting Ganga water for industry and agriculture. GAP also neglected non-point source pollutants like agricultural runoff.	Adaption of river basin approach – This is crucial because it addresses pollution in a broader sense and considers all sources of pollution in the basin.	Divergence of river action plan with broader development policies – The GAP's main goal is Ganga cleansing, hence it must be integrated with other development programmes like industrialisation, urbanisation, and irrigation. Without such integration, many of Ganga's fundamental pollution issues would remain unresolved.
Creation of institutional structure – Establishing wide institutional framework at national, state, and local levels was crucial to programme execution.	Inadequacy of standards of water quality – In the context of Ganga-bank religious ceremonies, 'bathing-class' is insufficient. Contrary to Western nations, where 'bathing-class' is sufficient as river water is seldom utilised for 'human-contact uses'.	Experiences with technologies – Government authorities comprehend the pros and cons of GAP treatment methods such as Anaerobic Sludge Blanket (UASB), Activated Sludge Process (ASP), and Stabilisation Pond Technology. This experience must be used with modern technology.	Challenge in experimentation with newer institutional models – Future engagement of private sector in Ganga Action Plan is GOI's policy. However, reform methods like the Public Private Partnership (PPP) model are still unproven as robust and effective enough and may fail if implemented broadly.
Strengths of operation and maintenance of STPs – Having diesel pump sets to run STPs during load-shedding schedules and power outages was a crucial step.	Inappropriate technological choices for treatment – Activated Sludge Processes (ASP), adopted in GAP is highly incapable in terms of removing pathogens and coliform.	GOIs commitment to raise adequate funds – The strategy described by NCRD in the future course of action assures approaching all possible agencies for raising financial resources such as, state governments, central government, etc.	Wastage of funds – Misuse and wastage of funds is one of the serious challenges in terms of budget preparation, cost-overruns and quality of the work done.
Monitoring through various stakeholders – GAP was monitored and reviewed through various stake holders like CGA, CPCB, SPCBs, NCRD, SRCDS and other ministerial agencies, as well as, District Collector.	Inappropriate policy of discharging water into the river – Discharging partially treated sewage or effluent in the river is permissible if (a) river flows are substantial and (b) treatment plants are operating at full capacity and diligence under effective regulation. Even when water flows are low for 10 months, the Indian model merely recommends secondary level treatment.	Awareness and inclination to contribute – Civil society organisations' awareness-building, analysis, and monitoring of government activities are unsurpassed. So, civil society organisations may help establish and manage future phases of GAP.	
Appointment of independent agencies for water quality monitoring – Many independent institutions, such as IIT Kanpur and University of Patna, monitored river water quality using CPCB/ NCRD methodology by collecting samples from different stretches of Ganga.	Weakness of monitoring, evaluation and regulation – Failure to use, available monitoring data for controlling industrial pollution; designing and establishing citizen monitoring committees as well as central institution monitoring.		

(Continued.)

Table 1 | Continued

Strengths of GAP	Weaknesses of GAP	Opportunities for future	Threats and challenges
Improvements in river water quality – The MoEF – GAP status report (NRCD 2009; prepared by AHEC, IIT Roorkee) recommends river water quality improvements against Menon Committee indicators, excluding coliform levels.	Lack of a clear policy-legal and institutional framework – The lack of clarity about the roles of various stakeholders involved in the implementation of the GAP.		

IMPACT OF COVID-19 PANDEMIC LOCKDOWN ON POLLUTION

The recent lockdown (March 25 till May 17, 2020) in India due to the coronavirus pandemic has given us a chance to rethink the cleaning strategies of the River Ganga, as its water quality was improved due to the nationwide lockdown. In a recent study, Dutta *et al.* reported that the health of the River Ganga was improved due to the lockdown (Dutta *et al.* 2020). It was found that the water quality at 27 monitoring stations was suitable for bathing (Class B) out of 36 monitoring stations placed on the mainstream of the River Ganga. Among these 36 monitoring stations, 9 stations were also suitable for the propagation of wildlife and fisheries (Class C). Accordingly, the improvement in water quality was attributed to the reduction in the release of industrial wastewater. This was further facilitated by excessive rainfall in the Gangetic plains during lockdown. However, BOD, COD, and organic loads were marginally improved, whereas the value of ammoniacal nitrogen was partially increased during the lockdown period. This was attributed to wastewater discharge from municipal sources as domestic discharges remained unchanged during the lockdown. However, limited industrial activities during the lockdown period resulted in improvement in nitrate concentration and bacteriological quality of the River Ganga. On contrary, Sen and co-workers emphasised that all the vital water quality parameters of River Ganga were not improved during the lockdown period and only the dissolved heavy metal load was improved (Shukla *et al.* 2021). Satellite-based (Sentinel-2) water quality analysis before and during lockdown was performed by Muduli *et al.* for seven selected locations spread across the entire stretch of the Ganges (Rishikesh–Dimond Harbour) (Muduli *et al.* 2021). It was found that the water quality of the Ganga improved with reference to specific water quality parameters, but the improvements were region-specific. For example, only the Haridwar site showed improvement to the extent of being potable as per the threshold set by the Central Pollution Control Board, New Delhi, India. The water quality of the River Ganga has been evaluated by Singh *et al.* during the lockdown and predicted for post-lockdown conditions (Singh *et al.* 2023). Lockdown resulted in the reduction of BOD ranging from 55 to 92% with increased concentration of dissolved oxygen at a few stations. pH was in the range of 6.5–8.5 during lockdown. The total coliform count declined during the lockdown period at some stations. Several researchers have also studied the impact of the COVID lockdown on river health and found the improvement was region-specific (Garga *et al.* 2020; Duttagupta *et al.* 2021; Singh & Pandey 2022). From these studies, it is clear that the problem of water quality deterioration of the River Ganga is mostly anthropogenic. From the way the water quality was improved during the lockdown period, it is evident that adequate flow of the river is crucial for dilution of the pollutants. Thus, apart from managing sewerage and industrial effluents, it is also equally important to rationalise irrigation and power projects in order to maintain the ecological flow of rivers. Furthermore, industries and residential projects should be restricted in river catchment areas. Similarly, mining and tourism should also be strictly rationalised for the good of the environment.

CONCLUSIONS

This review article summarises the factors responsible for pollution in the River Ganga and the threat to its biodiversity. It has been found that the discharge of sewage is the major cause of concern followed by industrial pollution. The concentrations of various toxic elements in water and sediments of rivers are above the acceptable limit. These toxicants are entering the environment through aquatic life and posing a high risk to human health due to their bioaccumulation and biomagnification. Hence, immediate actions are required to minimise the toxicant load into the River Ganga. Apart from sewage and industrial pollution, socio-economic activities are also posing high risks to the health of the river and its biodiversity. Construction of

dams, barrages, and bridges not only destroys the habitats of many animals but also restricts the migration of many aquatic animals. Irrigation and dams have reduced the water flow in rivers below the ecological flow, thereby risking the lives of many aquatic animals. Even religious activities severely pollute the river water due to mass bathing, idol immersion, and cremations. Government projects to clean the Ganga are mainly based on the current load and sources of pollution. It is important to estimate the capacity of STP based on an increase in wastewater generation in the Ganga basin in the coming decades. The improvement in the water quality of the River Ganga during the COVID-19 pandemic lockdown suggested that strict monitoring of pollution and reduction in its load is the topmost requirement for a clean Ganga. The level of the water flow in the River Ganga is far from its mandatory ecological flow. Ecological flow of the river is necessary for its self-cleaning characteristic and the survival of aquatic life. Therefore, the maintenance of the ecological flow of rivers must be a top priority. The government alone cannot clean the Ganga River, therefore, people's participation is necessary for cleaning the Ganga.

The purpose of this study was to investigate the complexity of the Ganga River system and its current status in connection to a variety of factors pertaining to the socio-economic environment. These elements have an effect on the river's physiochemical qualities as well as its ability to support biodiversity. In addition to evaluating the efficacy of rejuvenation programmes, an analysis was conducted to determine the potential obstacles that these programmes may encounter in the future when it comes to enhancing the overall condition of the Ganga river system. The study brought to light the dynamic aspect of riverine ecology and emphasised the adverse influence that human activities have on the quality of rivers. The report recommends reducing the amount of human involvement in the Ganga river ecosystem in order to ensure long-term conservation that is sustainable. There are some regions that have been recognised as crucial microenvironments that have an impact on the whole river system. These areas include dam sites, boat connections, overfishing zones, human usage areas, and pollutant dumping sites. These areas require careful attention in the planning activities that will be undertaken in the future. In spite of the fact that the Ganga river's terrain is very dynamic, with dense populations and strong commercial interests, it continues to be hydrographically complicated and susceptible. The study also found that during the COVID-19 pandemic, there was a considerable improvement in river conditions. This improvement was attributable to the fact that there was less intervention by humans into the natural ecology of the Ganga. In light of this, the study highlights the need to reduce the influence that the humans have on the Ganga river and implement rejuvenation programmes that are more successful in order to improve the river's state in the future.

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DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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